QUANTITATIVE METHODS 1

Worksheet 3 – Dispersion and concentration

Detailed solutions

Exercise 1 – Detailed solution

$$\min = 5, \qquad Me = 8, \qquad \max = 20.$$

Lower half $5, 6, 6, 7, 7, 8$ gives

$$Q_1 = \frac{6+7}{2} = 6.5.$$

Upper half $8, 9, 9, 10, 11, 20$ gives

$$Q_3 = \frac{9+10}{2} = 9.5.$$

Thus

$$IQR = 9.5 - 6.5 = 3.$$

The fences are

$$B_L = 6.5 - 1.5(3) = 2, \qquad B_U = 9.5 + 1.5(3) = 14.$$

Since $20 > 14$, 20 is a potential outlier.

$$\bar{x} = \frac{106}{12} \approx 8.8333.$$

$$MAD = \frac{1}{12} \sum |x_i - \bar{x}| \approx 2.4722.$$

Population variance and standard deviation:

$$\sigma^2 \approx 14.1389, \qquad \sigma \approx 3.7602.$$

Sample variance and standard deviation:

$$s^2 \approx 15.4242, \qquad s \approx 3.9274.$$

Exercise 2 – Detailed solution

$$\sum n_i c_i = 6665, \qquad \sum n_i c_i^2 = 227036.5.$$

$$\bar{x} = \frac{6665}{210} \approx 31.7381.$$

$$V = \frac{227036.5}{210} - \bar{x}^2 \approx 73.8195.$$

$$\sigma \approx 8.5918, \qquad CV = \frac{8.5918}{31.7381} \approx 27.07\%.$$

$$IQR = 37.50 - 26.56 \approx 10.94.$$

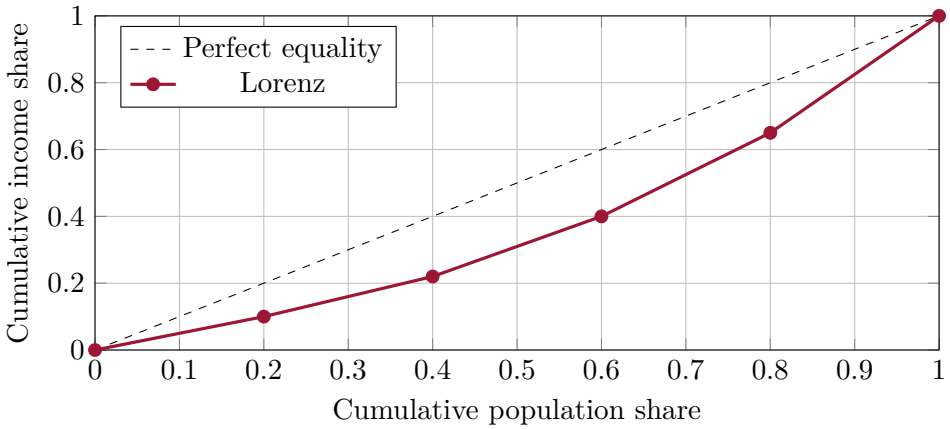
Standard deviation uses the full distribution and is sensitive to extremes; the IQR describes the spread of the central half and is more robust.

Exercise 3 – Detailed solution

Total income is 100. Lorenz points are

$$(0,0),\ (0.2,0.10),\ (0.4,0.22),\ (0.6,0.40),\ (0.8,0.65),\ (1,1).$$

$$A = 0.374, \qquad G = 1 - 2A = 0.252 = 25.2\%.$$



The Gini coefficient summarizes concentration; it does not identify the economic or institutional mechanisms causing it.